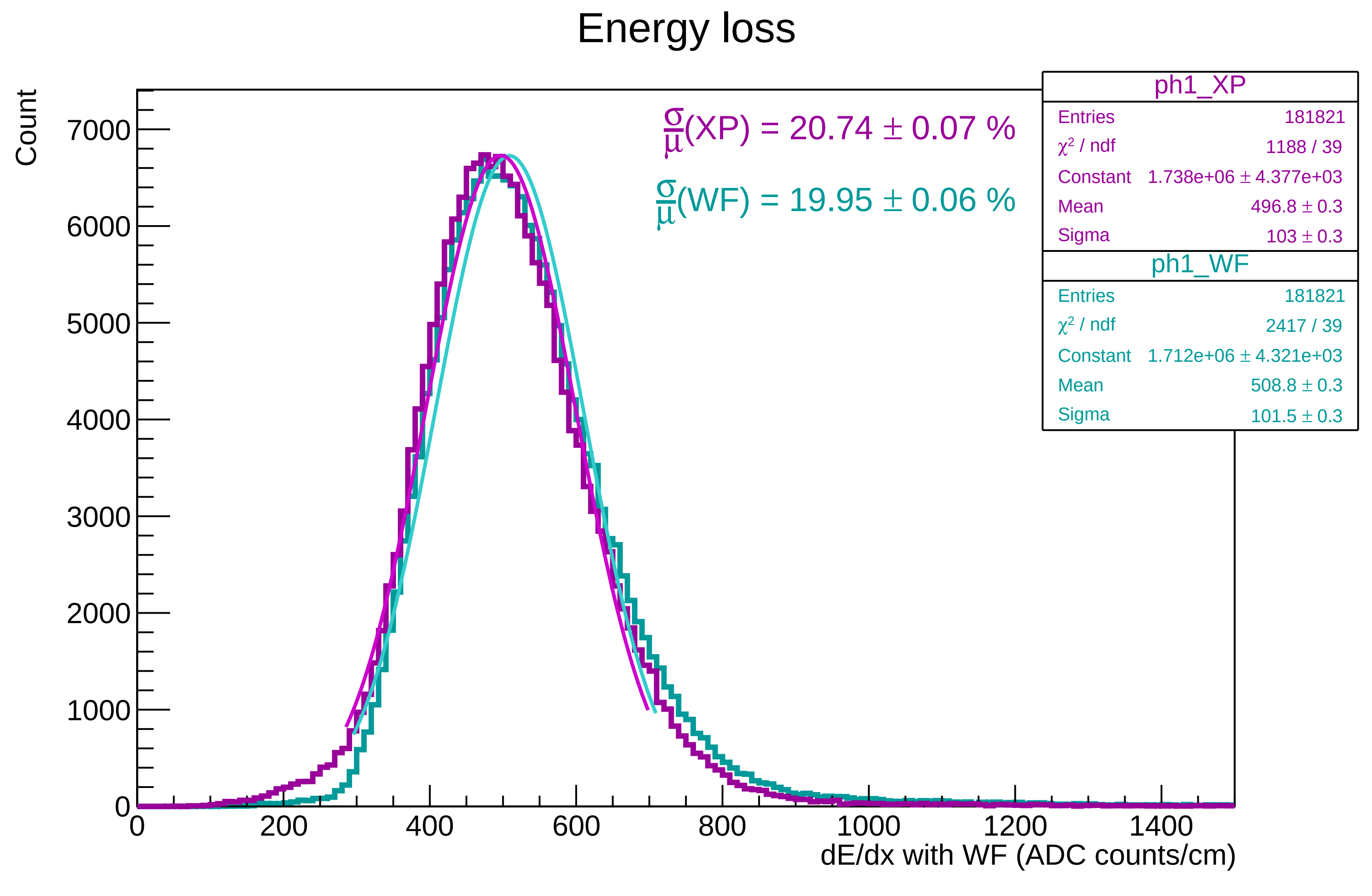
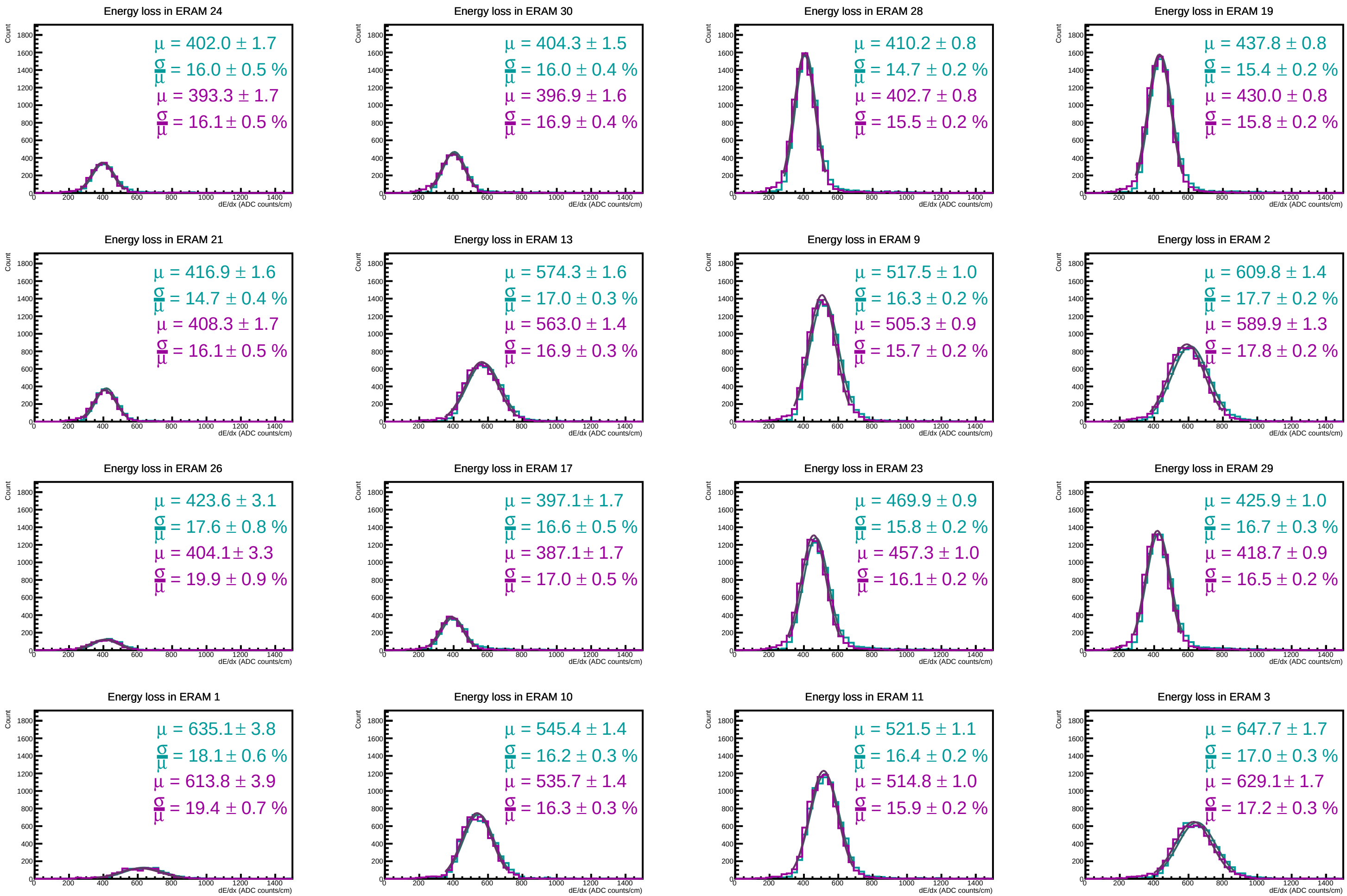
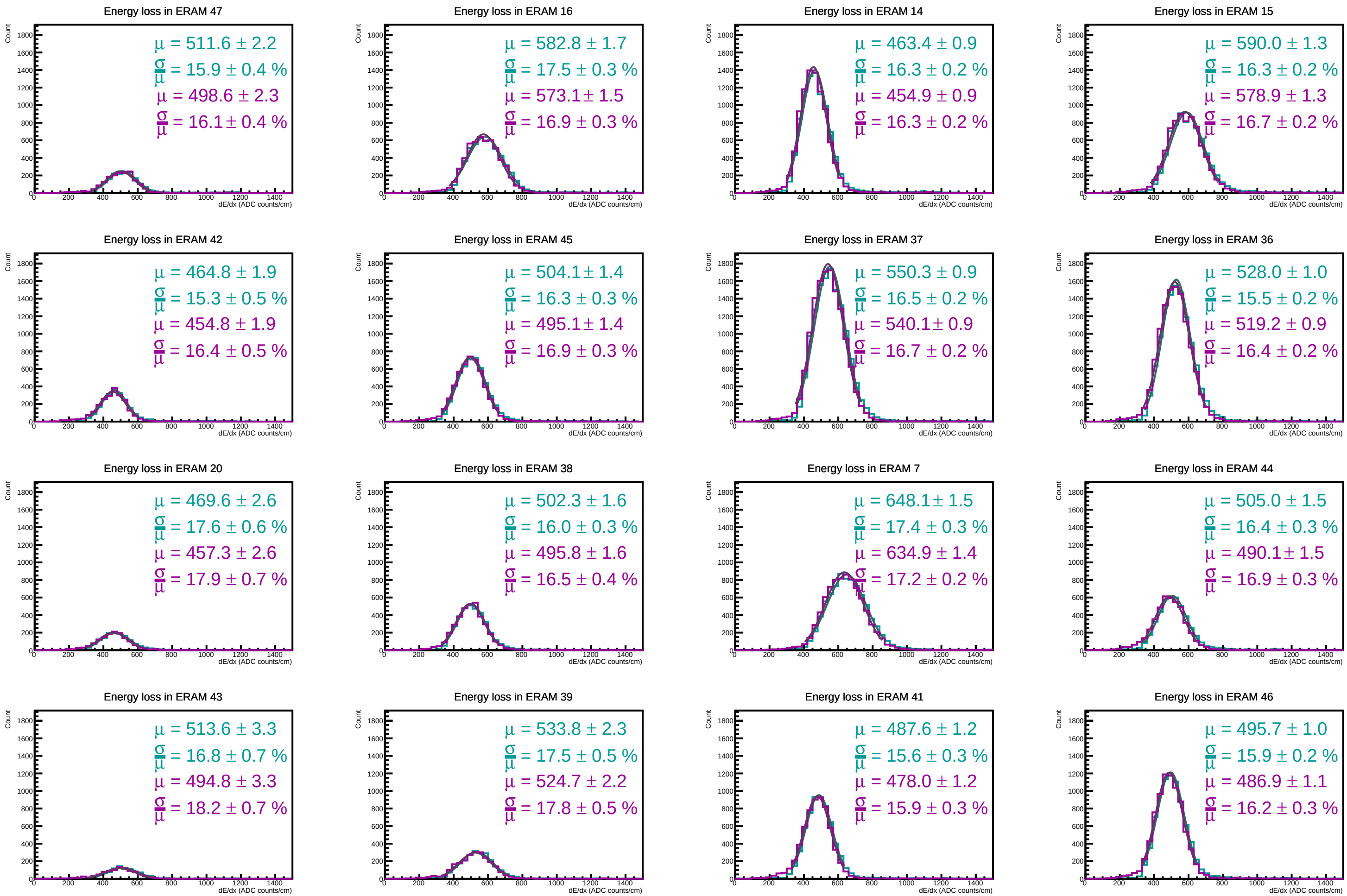
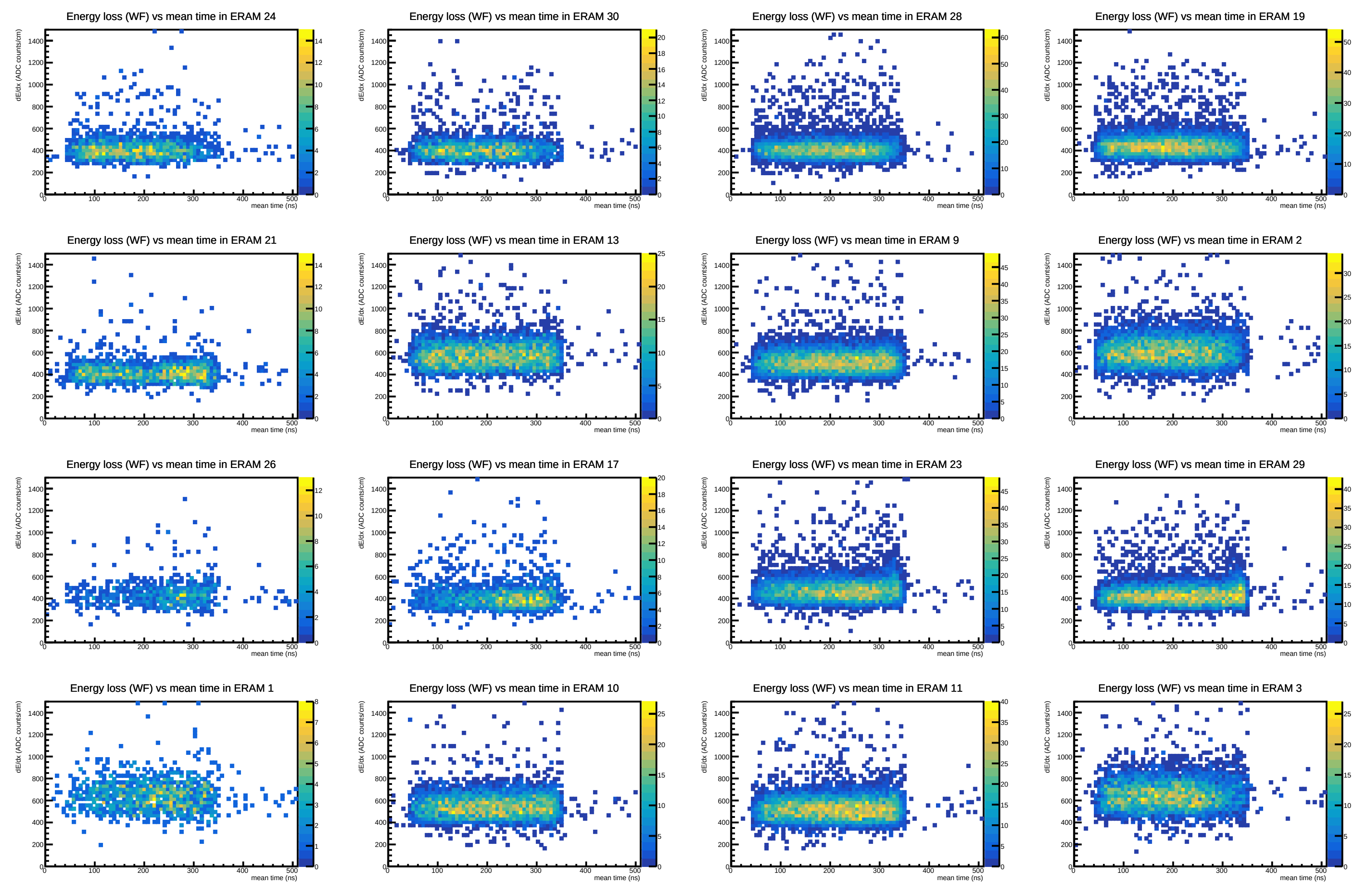
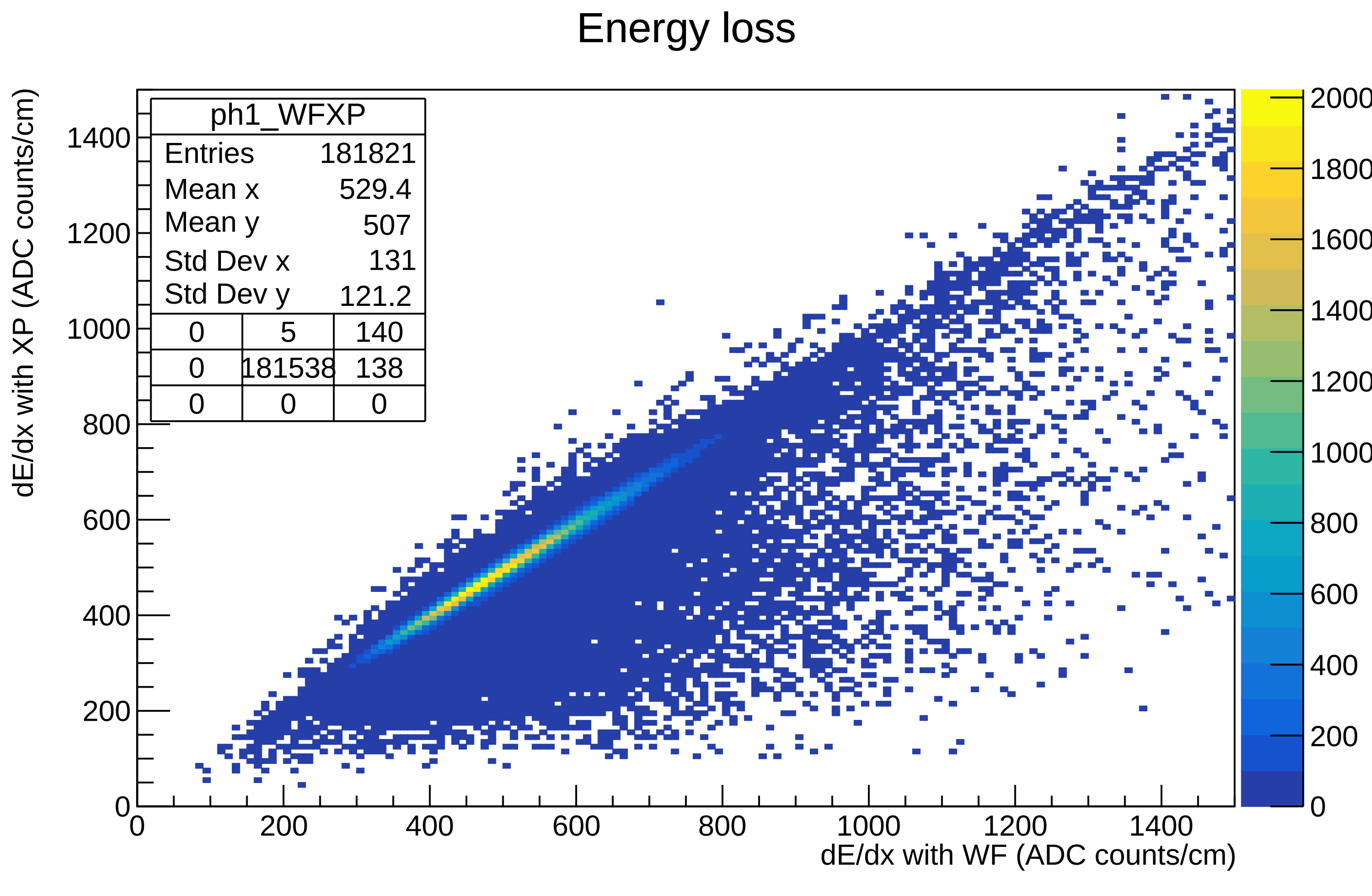

Energy loss
Count
dE/dx with WF (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
7000
0
200
400
600
800
1000
1200
1400
σ/μ(XP) = 20.74 ± 0.07 %
σ/μ(WF) = 19.95 ± 0.06 %
ph1_XP
Entries 181821
χ2 / ndf 1188 / 39
Constant 1.738e+06 ± 4.377e+03
Mean 496.8 ± 0.3
Sigma 103 ± 0.3
ph1_WF
Entries 181821
χ2 / ndf 2417 / 39
Constant 1.712e+06 ± 4.321e+03
Mean 508.8 ± 0.3
Sigma 101.5 ± 0.3

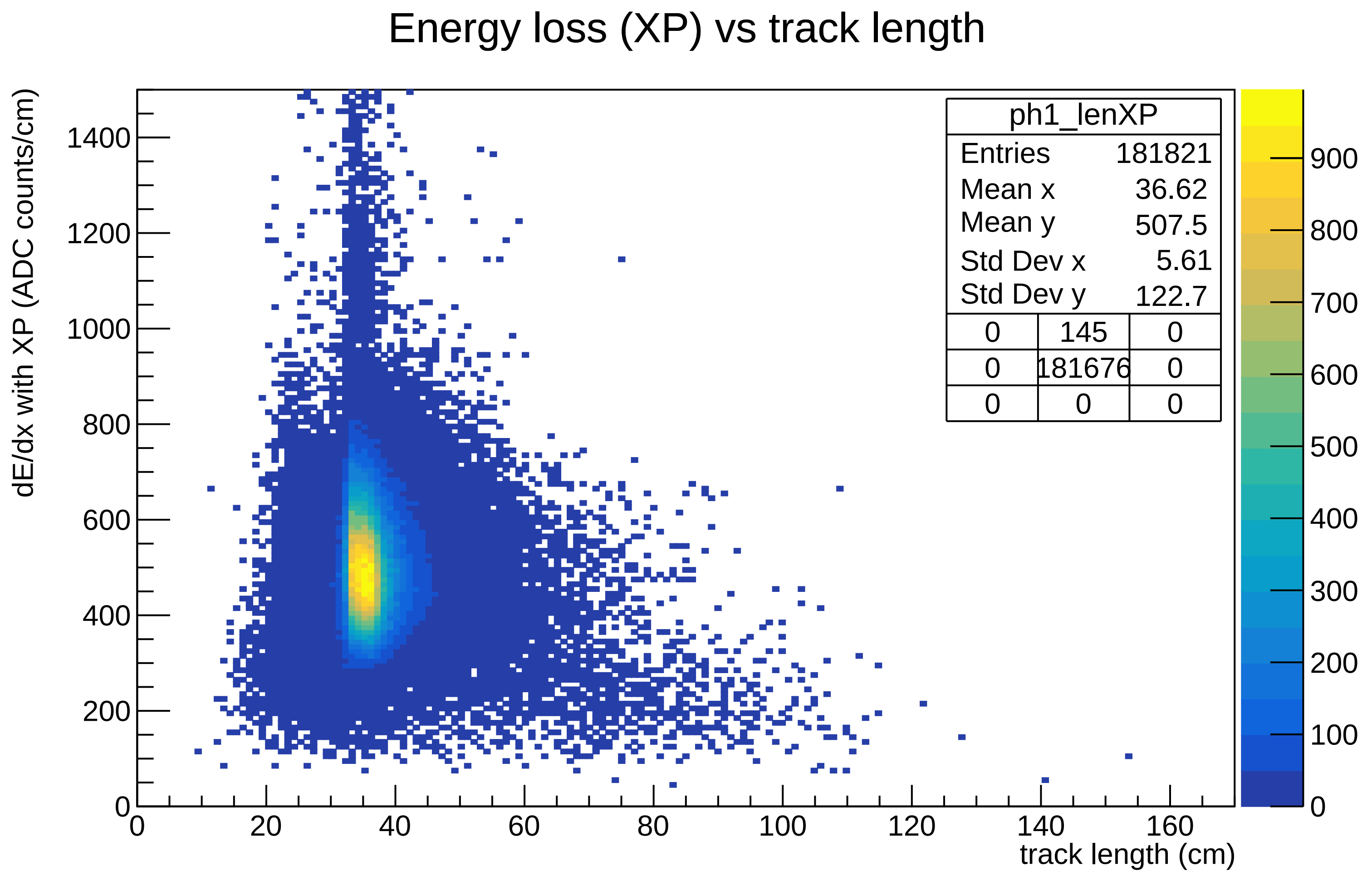

Energy loss in ERAM 24
μ = 402.0 ± 1.7
σ/μ = 16.0 ± 0.5 %
μ = 393.3 ± 1.7
σ/μ = 16.1 ± 0.5 %
Energy loss in ERAM 30
μ = 404.3 ± 1.5
σ/μ = 16.0 ± 0.4 %
μ = 396.9 ± 1.6
σ/μ = 16.9 ± 0.4 %
Energy loss in ERAM 28
μ = 410.2 ± 0.8
σ/μ = 14.7 ± 0.2 %
μ = 402.7 ± 0.8
σ/μ = 15.5 ± 0.2 %
Energy loss in ERAM 19
μ = 437.8 ± 0.8
σ/μ = 15.4 ± 0.2 %
μ = 430.0 ± 0.8
σ/μ = 15.8 ± 0.2 %
Energy loss in ERAM 21
μ = 416.9 ± 1.6
σ/μ = 14.7 ± 0.4 %
μ = 408.3 ± 1.7
σ/μ = 16.1 ± 0.5 %
Energy loss in ERAM 13
μ = 574.3 ± 1.6
σ/μ = 17.0 ± 0.3 %
μ = 563.0 ± 1.4
σ/μ = 16.9 ± 0.3 %
Energy loss in ERAM 9
μ = 517.5 ± 1.0
σ/μ = 16.3 ± 0.2 %
μ = 505.3 ± 0.9
σ/μ = 15.7 ± 0.2 %
Energy loss in ERAM 2
μ = 609.8 ± 1.4
σ/μ = 17.7 ± 0.2 %
μ = 589.9 ± 1.3
σ/μ = 17.8 ± 0.2 %
Energy loss in ERAM 26
μ = 423.6 ± 3.1
σ/μ = 17.6 ± 0.8 %
μ = 404.1 ± 3.3
σ/μ = 19.9 ± 0.9 %
Energy loss in ERAM 17
μ = 397.1 ± 1.7
σ/μ = 16.6 ± 0.5 %
μ = 387.1 ± 1.7
σ/μ = 17.0 ± 0.5 %
Energy loss in ERAM 23
μ = 469.9 ± 0.9
σ/μ = 15.8 ± 0.2 %
μ = 457.3 ± 1.0
σ/μ = 16.1 ± 0.2 %
Energy loss in ERAM 29
μ = 425.9 ± 1.0
σ/μ = 16.7 ± 0.3 %
μ = 418.7 ± 0.9
σ/μ = 16.5 ± 0.2 %
Energy loss in ERAM 1
μ = 635.1 ± 3.8
σ/μ = 18.1 ± 0.6 %
μ = 613.8 ± 3.9
σ/μ = 19.4 ± 0.7 %
Energy loss in ERAM 10
μ = 545.4 ± 1.4
σ/μ = 16.2 ± 0.3 %
μ = 535.7 ± 1.4
σ/μ = 16.3 ± 0.3 %
Energy loss in ERAM 11
μ = 521.5 ± 1.1
σ/μ = 16.4 ± 0.2 %
μ = 514.8 ± 1.0
σ/μ = 15.9 ± 0.2 %
Energy loss in ERAM 3
μ = 647.7 ± 1.7
σ/μ = 17.0 ± 0.3 %
μ = 629.1 ± 1.7
σ/μ = 17.2 ± 0.3 %
Count
dE/dx (ADC counts/cm)



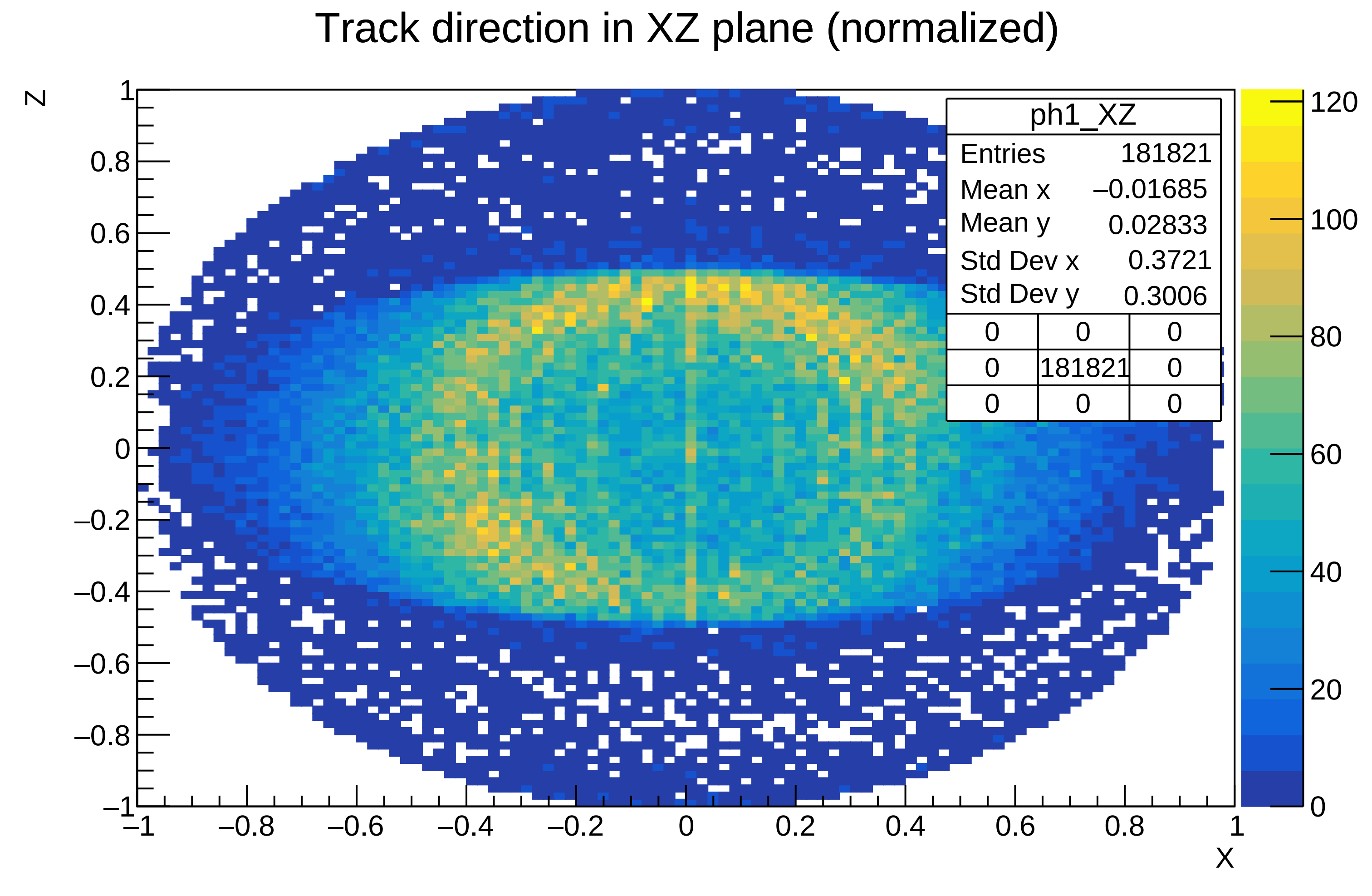
Energy loss in ERAM 47
μ = 511.6 ± 2.2
σ/μ = 15.9 ± 0.4 %
μ = 498.6 ± 2.3
σ/μ = 16.1 ± 0.4 %
Energy loss in ERAM 16
μ = 582.8 ± 1.7
σ/μ = 17.5 ± 0.3 %
μ = 573.1 ± 1.5
σ/μ = 16.9 ± 0.3 %
Energy loss in ERAM 14
μ = 463.4 ± 0.9
σ/μ = 16.3 ± 0.2 %
μ = 454.9 ± 0.9
σ/μ = 16.3 ± 0.2 %
Energy loss in ERAM 15
μ = 590.0 ± 1.3
σ/μ = 16.3 ± 0.2 %
μ = 578.9 ± 1.3
σ/μ = 16.7 ± 0.2 %
Energy loss in ERAM 42
μ = 464.8 ± 1.9
σ/μ = 15.3 ± 0.5 %
μ = 454.8 ± 1.9
σ/μ = 16.4 ± 0.5 %
Energy loss in ERAM 45
μ = 504.1 ± 1.4
σ/μ = 16.3 ± 0.3 %
μ = 495.1 ± 1.4
σ/μ = 16.9 ± 0.3 %
Energy loss in ERAM 37
μ = 550.3 ± 0.9
σ/μ = 16.5 ± 0.2 %
μ = 540.1 ± 0.9
σ/μ = 16.7 ± 0.2 %
Energy loss in ERAM 36
μ = 528.0 ± 1.0
σ/μ = 15.5 ± 0.2 %
μ = 519.2 ± 0.9
σ/μ = 16.4 ± 0.2 %
Energy loss in ERAM 20
μ = 469.6 ± 2.6
σ/μ = 17.6 ± 0.6 %
μ = 457.3 ± 2.6
σ/μ = 17.9 ± 0.7 %
Energy loss in ERAM 38
μ = 502.3 ± 1.6
σ/μ = 16.0 ± 0.3 %
μ = 495.8 ± 1.6
σ/μ = 16.5 ± 0.4 %
Energy loss in ERAM 7
μ = 648.1 ± 1.5
σ/μ = 17.4 ± 0.3 %
μ = 634.9 ± 1.4
σ/μ = 17.2 ± 0.2 %
Energy loss in ERAM 44
μ = 505.0 ± 1.5
σ/μ = 16.4 ± 0.3 %
μ = 490.1 ± 1.5
σ/μ = 16.9 ± 0.3 %
Energy loss in ERAM 43
μ = 513.6 ± 3.3
σ/μ = 16.8 ± 0.7 %
μ = 494.8 ± 3.3
σ/μ = 18.2 ± 0.7 %
Energy loss in ERAM 39
μ = 533.8 ± 2.3
σ/μ = 17.5 ± 0.5 %
μ = 524.7 ± 2.2
σ/μ = 17.8 ± 0.5 %
Energy loss in ERAM 41
μ = 487.6 ± 1.2
σ/μ = 15.6 ± 0.3 %
μ = 478.0 ± 1.2
σ/μ = 15.9 ± 0.3 %
Energy loss in ERAM 46
μ = 495.7 ± 1.0
σ/μ = 15.9 ± 0.2 %
μ = 486.9 ± 1.1
σ/μ = 16.2 ± 0.3 %
Count
dE/dx (ADC counts/cm)


Energy loss (WF) vs mean time in ERAM 24
Energy loss (WF) vs mean time in ERAM 30
Energy loss (WF) vs mean time in ERAM 28
Energy loss (WF) vs mean time in ERAM 19
Energy loss (WF) vs mean time in ERAM 21
Energy loss (WF) vs mean time in ERAM 13
Energy loss (WF) vs mean time in ERAM 9
Energy loss (WF) vs mean time in ERAM 2
Energy loss (WF) vs mean time in ERAM 26
Energy loss (WF) vs mean time in ERAM 17
Energy loss (WF) vs mean time in ERAM 23
Energy loss (WF) vs mean time in ERAM 29
Energy loss (WF) vs mean time in ERAM 1
Energy loss (WF) vs mean time in ERAM 10
Energy loss (WF) vs mean time in ERAM 11
Energy loss (WF) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

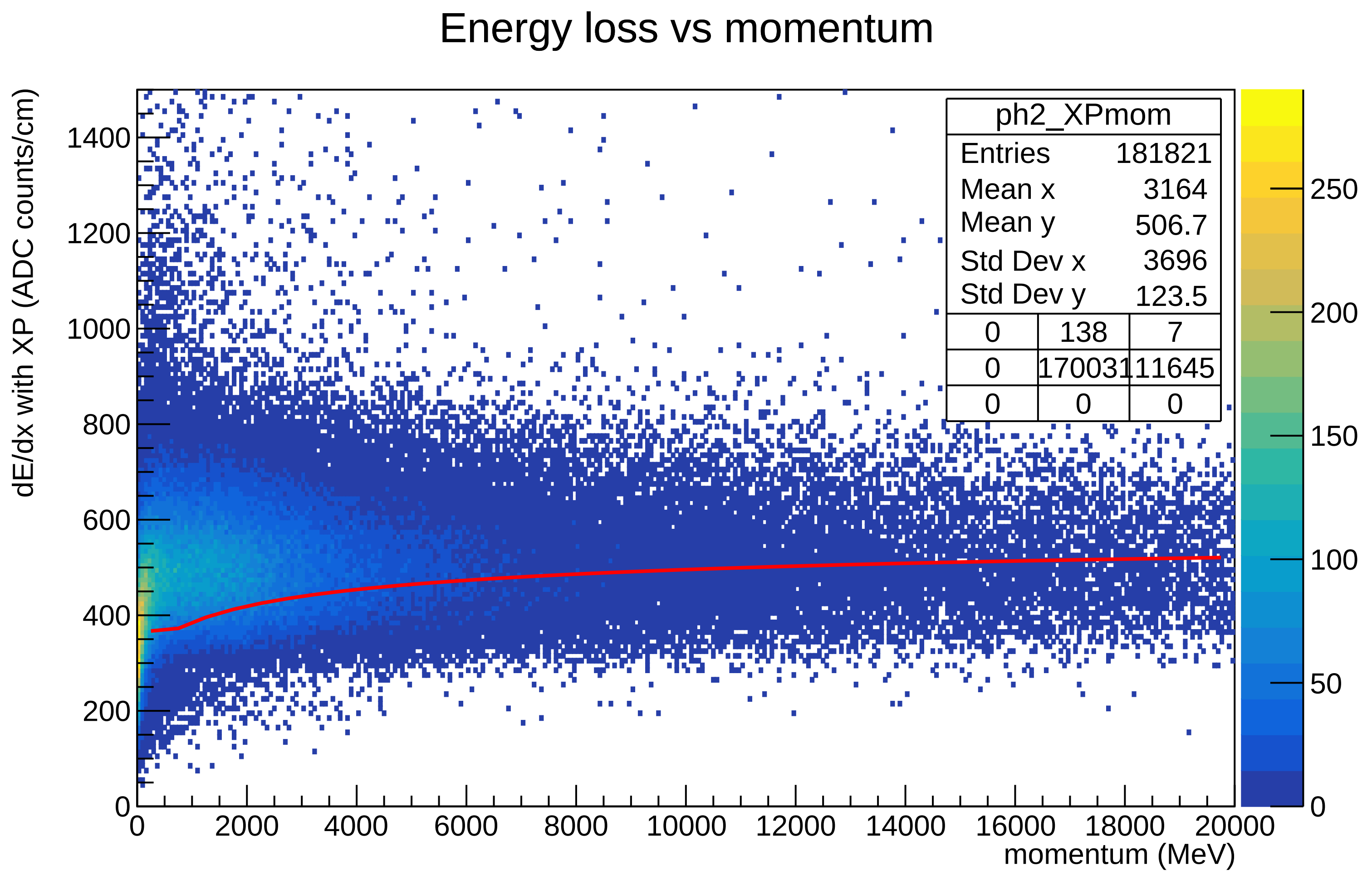

Energy loss
ph1_WFXP
Entries 181821
Mean x 529.4
Mean y 507
Std Dev x 131
Std Dev y 121.2
0 5 140
0 181538 138
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
1400
1600
1800
2000

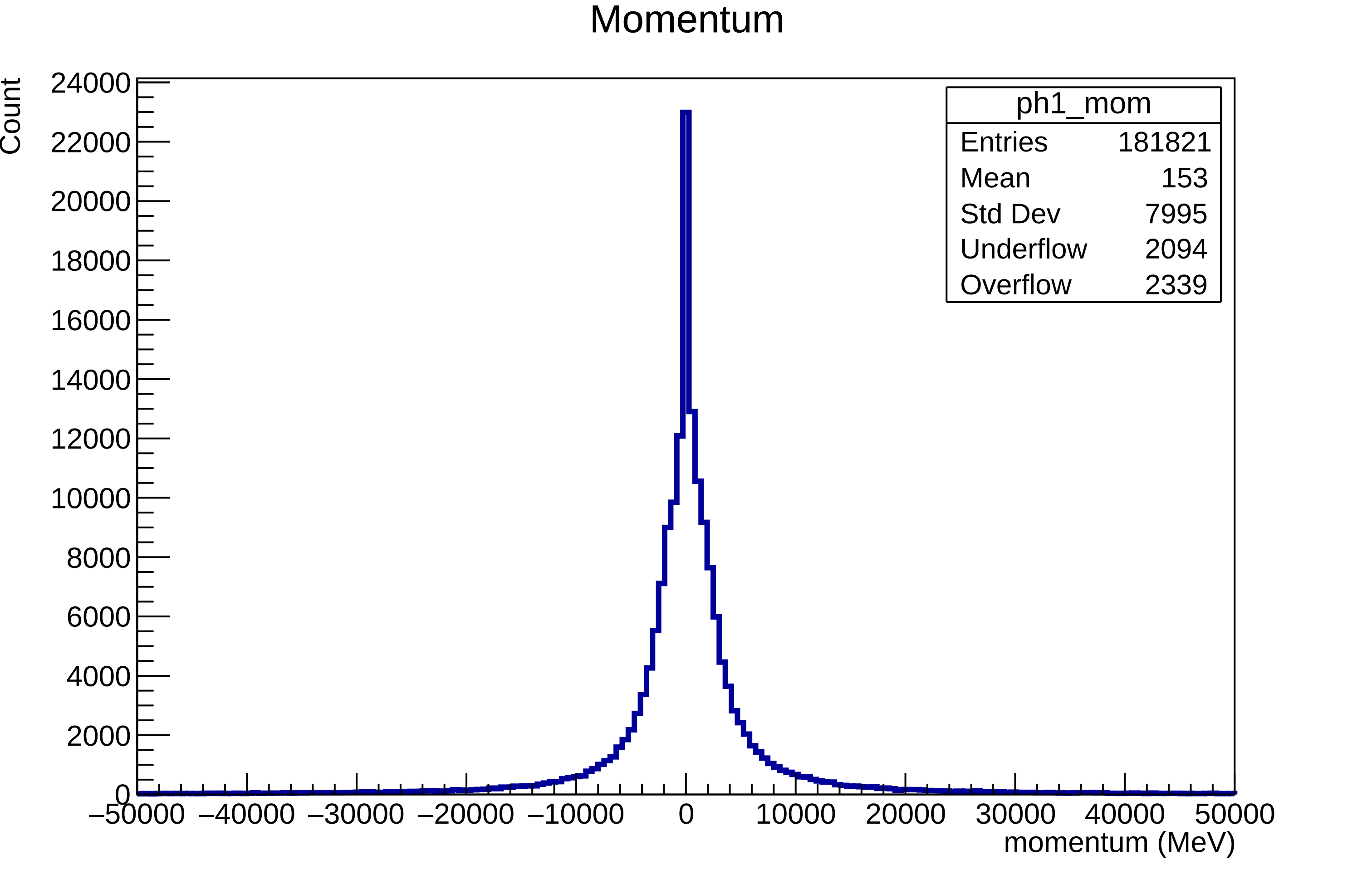


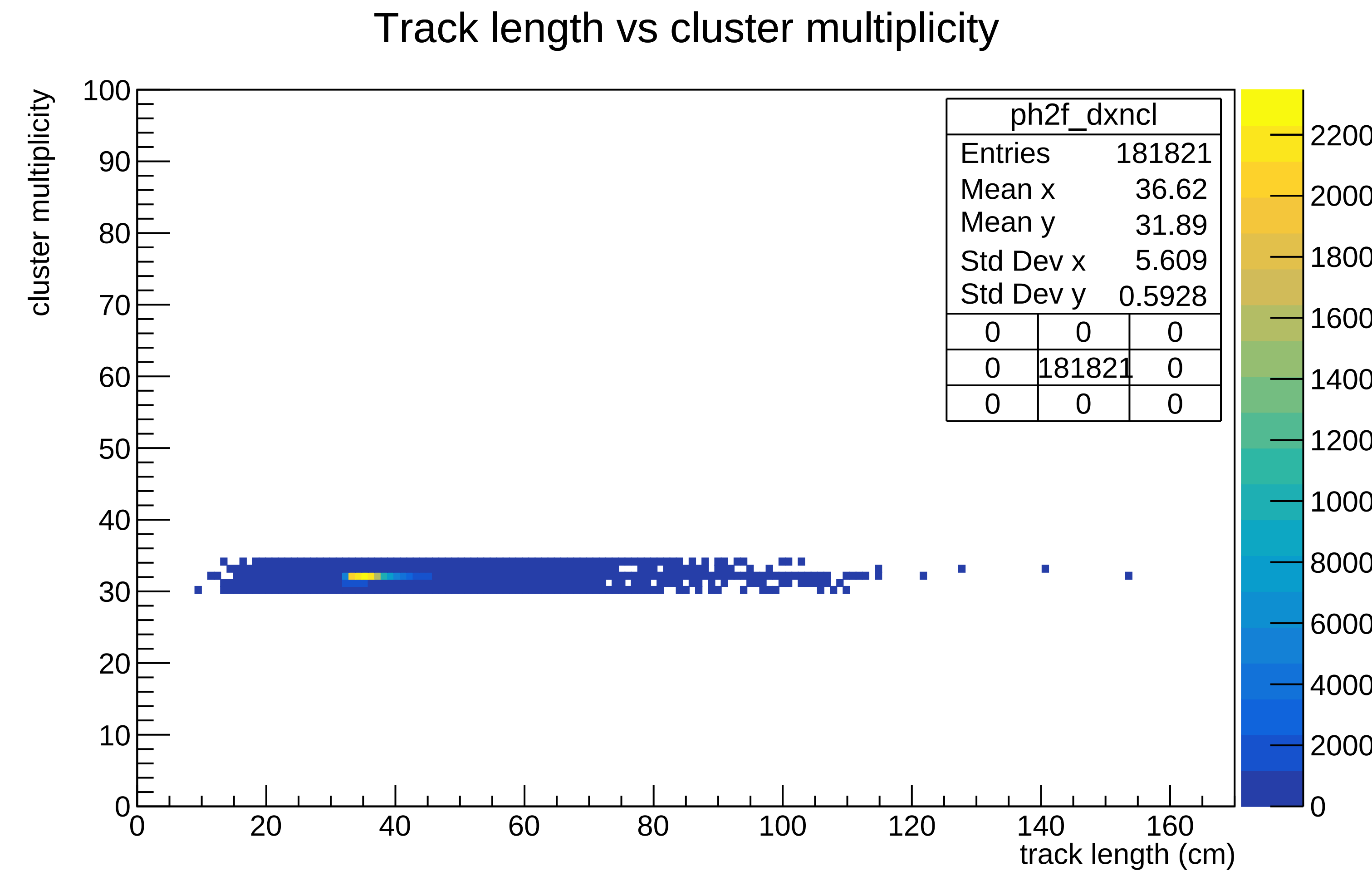
Energy loss (XP) vs track length
ph1_lenXP
Entries 181821
Mean x 36.62
Mean y 507.5
Std Dev x 5.61
Std Dev y 122.7
0 145 0
0 181676 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
200
400
600
800
1000
1200
1400
100
300
500
700
900

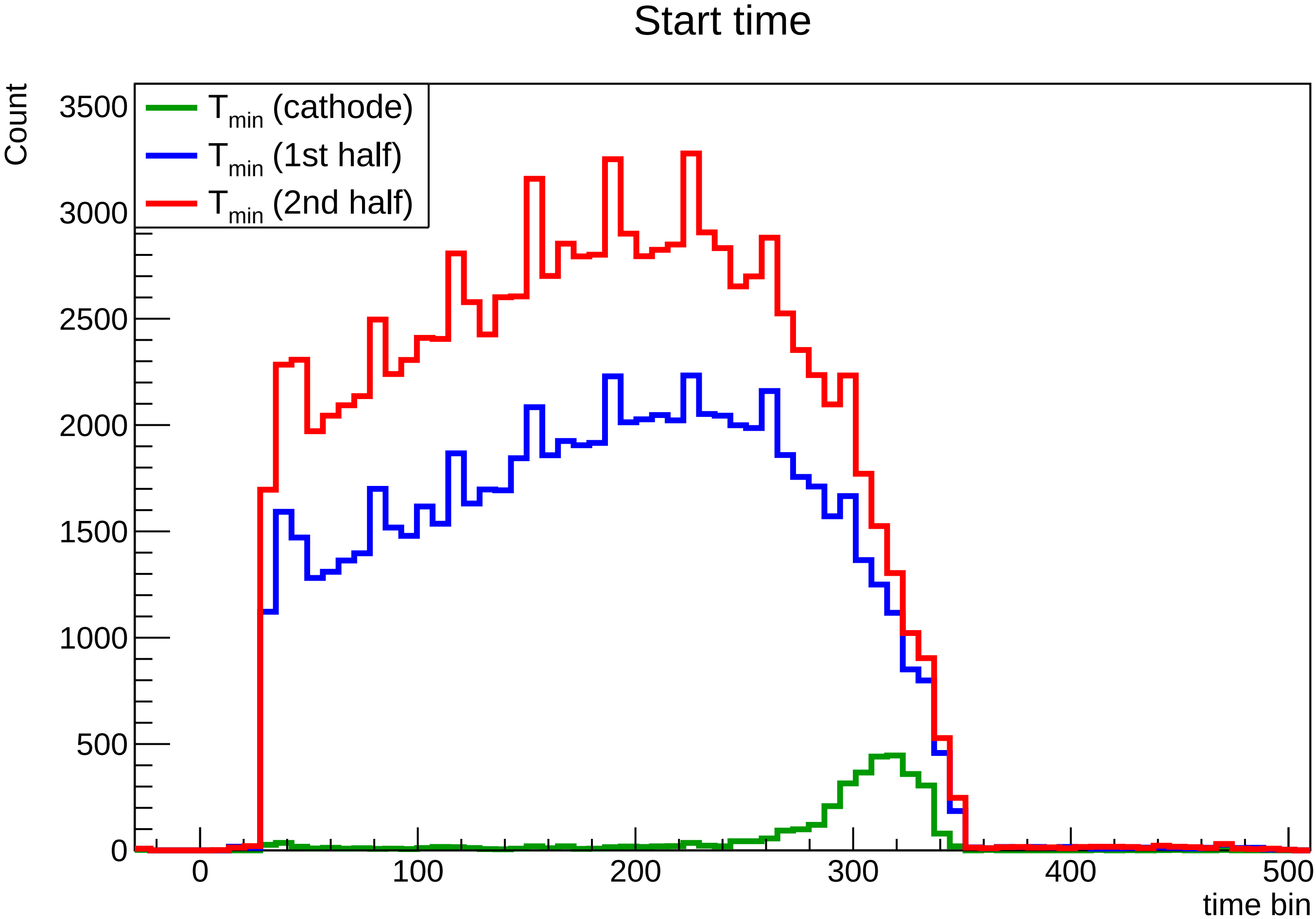


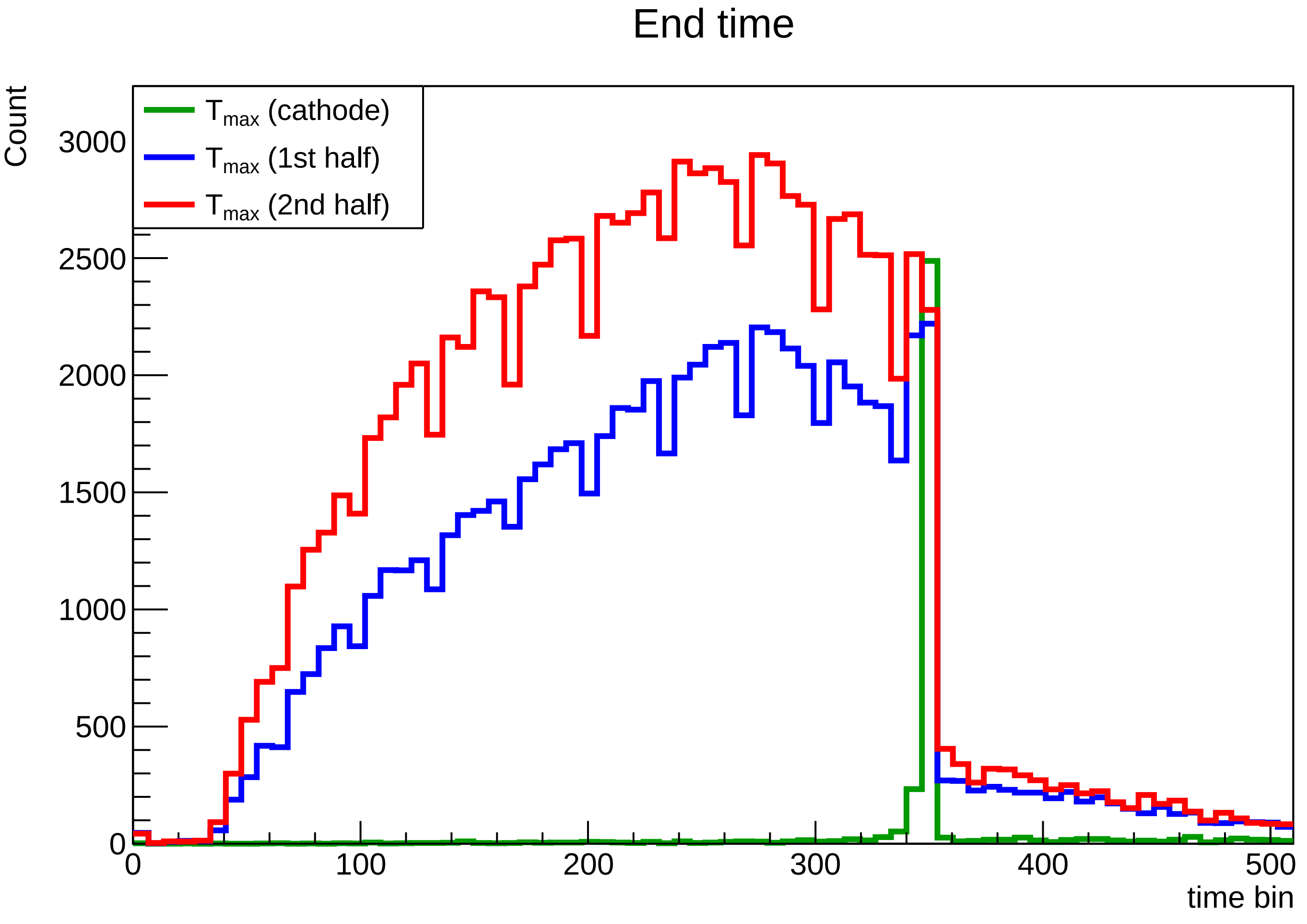
Track direction in XZ plane (normalized)
Z
X
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1
0
20
40
60
80
100
120
ph1_XZ
Entries 181821
Mean x −0.01685
Mean y 0.02833
Std Dev x 0.3721
Std Dev y 0.3006
0 0 0
0 181821 0
0 0 0


Energy loss vs momentum
ph2_XPmom
Entries 181821
Mean x 3164
Mean y 506.7
Std Dev x 3696
Std Dev y 123.5
0 138 7
0 170031 11645
0 0 0
dE/dx with XP (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
momentum (MeV)
0
50
100
150
200
250


Momentum
Count
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
–50000
–40000
–30000
–20000
–10000
0
10000
20000
30000
40000
50000
momentum (MeV)
ph1_mom
Entries 181821
Mean 153
Std Dev 7995
Underflow 2094
Overflow 2339


Track length vs cluster multiplicity
ph2f_dxncl
Entries 181821
Mean x 36.62
Mean y 31.89
Std Dev x 5.609
Std Dev y 0.5928
0 0 0
0 181821 0
0 0 0
cluster multiplicity
track length (cm)
0
20
40
60
80
100
120
140
160
10
30
50
70
90
2200
2000
1800
1600
1400
1200
1000
8000
6000
4000
2000


Start time
Count
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin
T min (cathode)
T min (1st half)
T min (2nd half)


End time
T_{max} (cathode)
T_{max} (1st half)
T_{max} (2nd half)
Count
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin